Mr. SQUID®: A Tabletop Platform for Transport, Flux Modulation, and Josephson Effects in a $YBa_2Cu_3O_{7-\delta}$ dc SQUID

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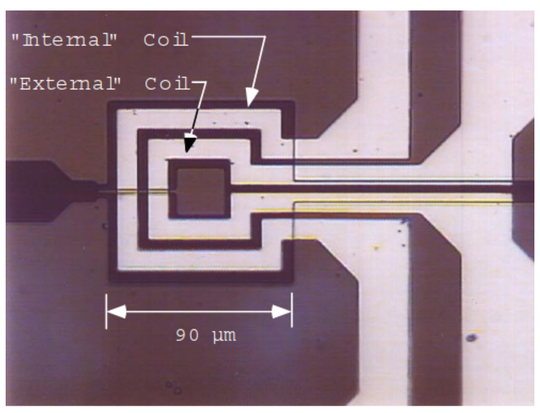
I. Aim

This work presents an experimental study of a high-T_c $YBa_2Cu_3O_{7-\delta}$ (YBCO) dc SQUID using the Mr. SQUID® tabletop platform [STAR Cryoelectronics, LLC(2000)]. The aim is to characterize the device through its temperature-dependent transition, Josephson transport, flux-periodic voltage modulation, and microwave-induced ac Josephson response.

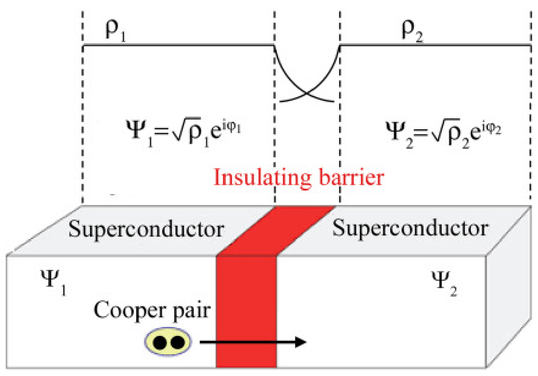
V–I characteristics $\longrightarrow$ V–Φ modulation $\longrightarrow$ R–T transition $\longrightarrow$ Shapiro steps

II. Device: high-T_c YBCO dc SQUID

The active element is a $YBa_2Cu_3O_{7-\delta}$ dc SQUID operated near liquid-nitrogen temperature. A dc SQUID is formed by placing two Josephson junctions in a superconducting loop. The phase difference across the junctions depends on the magnetic flux Φ threading the loop, producing a flux-periodic voltage response.



(a) dc SQUID loop with two Josephson junctions.



(b) Josephson tunneling across an insulating barrier [Vettoliere *et al.*(2023)Vettoliere, Silvestrini, and Granata].

Figure 1. Device principle: Josephson tunneling provides phase-sensitive weak links, while the SQUID loop converts magnetic flux into a measurable voltage modulation.

III. Experimental procedure

The same SQUID platform was operated in different modes to probe transport, flux response, transition behavior, and microwave-driven Josephson effects.

cooling $\rightarrow$ V–I $\rightarrow$ V–Φ $\rightarrow$ R–T $\rightarrow$ microwave drive

- **Cooling:** The SQUID probe was immersed in liquid nitrogen and stabilized near 77 K.
- **V–I:** Bias current was swept to identify the zero-voltage Josephson branch and resistive regions.
- **V–Φ:** The modulation-coil current was swept to observe flux-periodic voltage modulation.
- **R–T:** A calibrated silicon diode was used while the probe warmed through the superconducting transition.
- **Shapiro steps:** Microwave radiation was applied while recording driven V–I curves.

IV. Theory I: from Cooper pairing to SQUID interference

A dc SQUID works because a superconductor has a coherent macroscopic phase. In BCS theory, electrons in time-reversed states form Cooper pairs through an effective attractive interaction [Tinkham(1996)],

$$H_{\mathrm{BCS}} = \sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}}\, c^{\dagger}_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma} - \sum_{\mathbf{k},\mathbf{k}'} V_{\mathbf{k}\mathbf{k}'} c^{\dagger}_{\mathbf{k}\uparrow} c^{\dagger}_{-\mathbf{k}\downarrow} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}$$

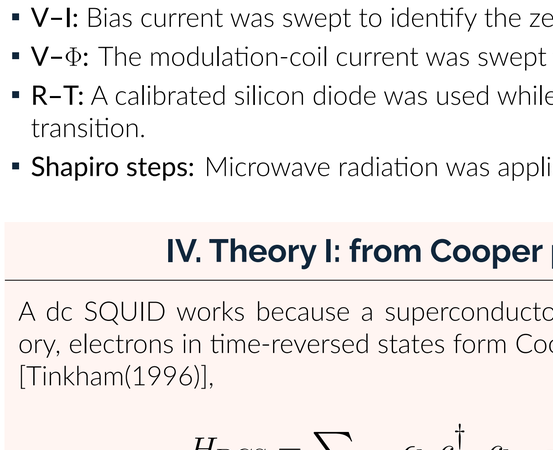
The paired condensate is described by : $\Psi(\mathbf{r}) = \sqrt{n_s(\mathbf{r})}\, e^{i\theta(\mathbf{r})}, \qquad \Delta_{\mathbf{k}} = -\sum_{\mathbf{k}'} V_{\mathbf{k}\mathbf{k}'} \langle c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow} \rangle.$
The phase θ controls Josephson tunneling. For a junction, $I_s = I_c \sin\Delta\theta, \qquad \frac{d\Delta\theta}{dt} = \frac{2eV}{\hbar}.$
In a SQUID loop, the phase differences across the two junctions are constrained by flux quantization,

$$\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n, \qquad \Phi_0 = \frac{h}{2e}.$$

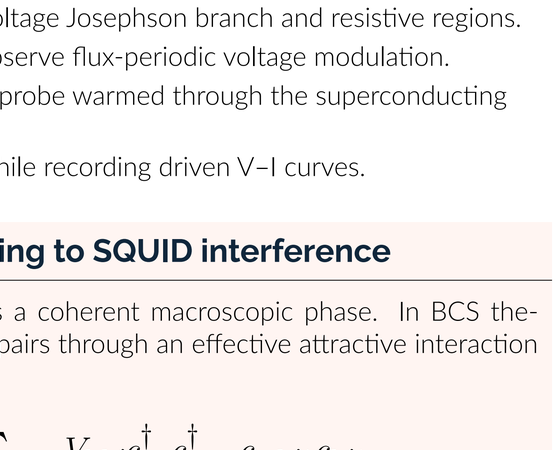
For an ideal symmetric dc SQUID, $I_{c,\mathrm{SQUID}}(\Phi) = 2I_c \left|\cos\left(\frac{\pi\Phi}{\Phi_0}\right)\right|$.

BCS pairing $\Rightarrow$ phase coherence $\Rightarrow$ Josephson current $\Rightarrow$ Φ_0-periodic $V(\Phi)$

V. Open-loop characterization: V–I transport and V–Φ interference



(a) V–I characteristic at 77 K; extracted $I_c = 27.520 \pm 0.177\ \mu\mathrm{A}$ and $R_N = 0.81244 \pm 0.00108\ \Omega$.



(b) V–Φ modulation curve with period $0.9943 \pm 0.0001\ \Phi_0$.

Figure 2. Open-loop SQUID response: V–I identifies the Josephson branch and resistive state, while V–Φ shows flux-periodic voltage modulation.

The V–I trace shows a zero-voltage Josephson branch and resistive branches at higher bias. Sweeping the modulation coil then gives a periodic V–Φ response with $V_{\Phi}^{\max} = 102.58 \pm 0.14\ \mathrm{mV}/\Phi_0$, confirming flux interference between the two junction paths.

VI. Theory II: superconducting transition and microwave locking

Superconducting transition

Near T_c, the Ginzburg–Landau free energy density is

$f = f_n + \alpha(T)|\Psi|^2 + \frac{\beta}{2}|\Psi|^4, \qquad \alpha(T) = \alpha_0(T - T_c).$
Minimization gives

$$|\Psi|^2 = \begin{cases} 0, & T > T_c, \\ \dfrac{\alpha_0(T_c - T)}{\beta}, & T < T_c. \end{cases}$$

Thus below T_c, $|\Psi|^2 \neq 0, n_s \propto |\Psi|^2, R(T) \to 0$.
The superconducting gap closes at the transition, $\Delta(T) \to 0 \qquad (T \to T_c^-)$.

Shapiro steps

Using the ac Josephson frequency,

$$f_J = \frac{2e}{h} V = \frac{V}{\Phi_0},$$

microwave phase locking occurs when

$$f_J = nf, \qquad n = 0, \pm 1, \pm 2, \ldots$$

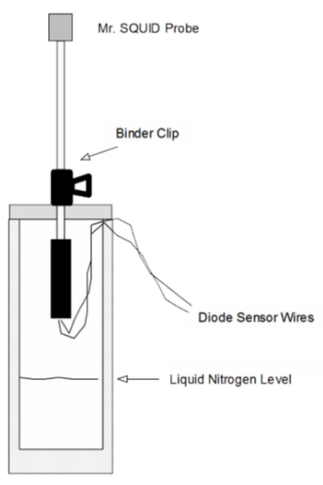
Therefore the voltage plateaus appear at

$$V_n = n\frac{hf}{2e} = n\Phi_0 f.$$

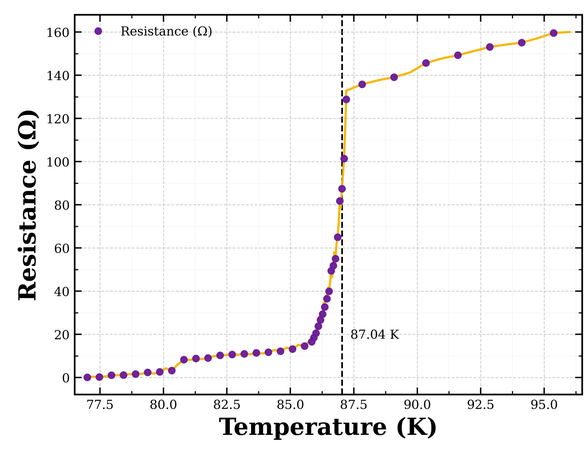
The step spacing is

$$\Delta V = V_{n+1} - V_n = \Phi_0 f.$$

VII. R–T measurement: superconducting transition



(a) R–T measurement setup with SQUID probe, diode sensor, and liquid nitrogen bath.

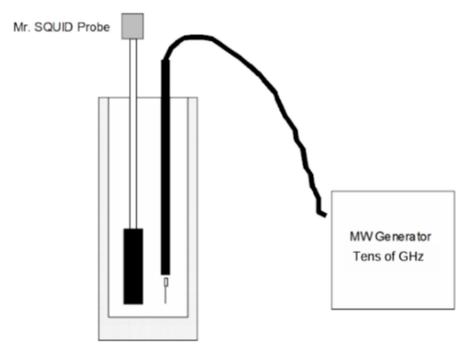


(b) Resistance–temperature curve of the YBCO SQUID device.

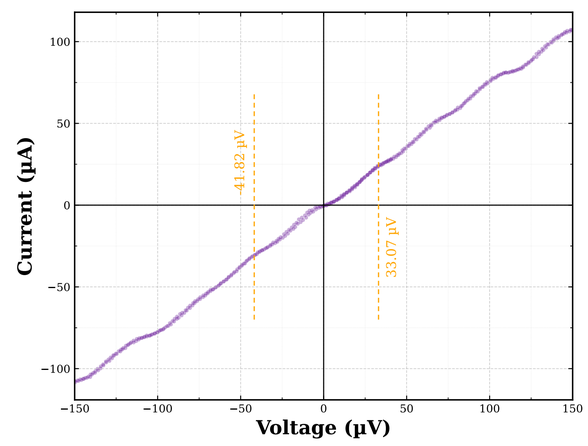
Figure 3. R–T measurement: the calibrated diode sensor tracks temperature while the SQUID probe warms through the superconducting transition.

The SQUID resistance was tracked while the probe warmed through the transition region. The sharp resistance drop ($T_c = 87.04 \pm 0.10$ K) confirms superconductivity above liquid-nitrogen temperature, consistent with high-T_c YBCO operation.

VIII. Shapiro steps: ac Josephson effect



(a) Microwave coupling setup for Shapiro-step measurement.



(b) Microwave-driven I–V curve showing Shapiro-step features.

Figure 4. Shapiro-step measurement: microwave radiation is coupled to the SQUID while recording the driven I–V response.

Under microwave irradiation, clear step-like plateaus are visible in the driven V–I curve, indicating phase locking of the Josephson oscillation to the external drive. For the applied frequency $f = 13.04$ GHz, the ideal first-step voltage is $V_{1,\mathrm{th}} = \Phi_0 f \approx 26.97\ \mu$V, while the observed average value was $V_{1,\mathrm{obs}} = 37.45 \pm 0.71\ \mu$V. The difference is likely due to broad and asymmetric step edges, finite readout resolution, and imperfect microwave coupling to the SQUID.

IX. Conclusion

The Mr. SQUID® system provides a compact tabletop demonstration of high-T_c dc SQUID physics. Using a single YBCO device, the experiment connects superconducting transition, Josephson transport, flux-periodic interference, and microwave-induced Shapiro steps.

Macroscopic superconducting phase coherence becomes directly measurable through transport, flux modulation, and microwave response.

X. Acknowledgements

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XI. References

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[Tinkham(1996)] M. Tinkham, *Introduction to Superconductivity*, 2nd ed. (McGraw-Hill, New York, 1996).